Freight Rate Prediction

Machine Learning Engineer assessment

Summary

The model predicts the posted rate of a load from its lane, distance, equipment, weight, date and the two market signals. It is validated with a **time-based, rolling-origin backtest**: each fold trains on everything before a cutoff and predicts the next two months, which mirrors the real task (train on Jan-Oct 2025, predict Nov-Dec 2025). Averaged over three folds, the final model's error is **2.31% MAPE** (MAE $56) on correctly labeled loads, against 3.43% for off-the-shelf gradient boosting and 3.71% for a rate-per-mile baseline.

1. Train/test split and validation approach

Why not a random split. The labeled data covers 1 Jan to 31 Oct 2025, and the loads to predict cover 1 Nov to 31 Dec 2025, entirely in the future. Rates depend on a daily market level and on a slow upward drift, so a random split would put loads from the same days and weeks in both train and test. That leaks the market conditions of the test period into training and overstates accuracy. The split therefore follows time.

Rolling-origin backtest. Three folds, each with a two-month test window (the same horizon as the real validation set) and an expanding training window:

Fold	Train on	Test on	Train rows	Test rows
1	2025-01-01 to 2025-07-01 (exclusive)	2025-07-01 to 2025-09-01 (exclusive)	28,806	9,671
2	2025-01-01 to 2025-08-01 (exclusive)	2025-08-01 to 2025-10-01 (exclusive)	33,718	9,429
3	2025-01-01 to 2025-09-01 (exclusive)	2025-09-01 to 2025-11-01 (exclusive)	38,477	9,523

Leakage controls, applied inside every fold:

- Everything learned from labels is fitted on the training window only: the corrupted-label filter, the log-linear coefficients and time trend, the lane encodings and the boosted trees.
- Lane encodings in the training window are computed **out-of-fold** (5-fold), so a row never sees its own label through its lane average.
- The test window's *unlabeled* features may be used to estimate each day's market level (the daily mean of market_index). This matches the real setting, where validation.csv provides market_index for every Nov-Dec load. Test labels are never used.
- After model selection, the final model is refitted on all of Jan-Oct with the same pipeline and hyperparameters, and then predicts validation.csv once.

Metrics. MAPE is reported as the headline metric because rates range from about $60 to $25,000 and a relative error is comparable across short and long hauls. MAE and RMSE are reported alongside it. Test labels flagged as corrupted (about 1.4% per fold, see section 2) are scored separately, because they reflect data-entry errors rather than model quality. Metrics on all rows are in the table in section 4.

2. Data validation and quality issues

Issue	Evidence	Handling
Corrupted rates (667 rows, 1.4%)	Rates at 0.2-0.5x or 2.5-4.6x the lane-equipment median, separated from genuine noise (~5%) by an almost empty gap. Occurrence is flat across equipment, month, weekday, distance and city, so the errors cannot be predicted.	Rows more than 1.6x from the structural model's estimate are excluded from training.
Negative weights (292 train, 145 validation)	Absolute values follow the same distribution as positive weights (median 31.8k vs 31.4k lb).	Treated as sign flips: abs().
Missing weight (300 / 165)	Under 1% of rows.	Median fill plus a missing-value flag.
Missing market_index (374 / 249)	market_index is a daily factor: std 0.025 within a day vs 0.167 between days.	Filled with that day's mean market_index.
8 unseen cities in validation (1,447 rows)	Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, San Diego.	Geography enters through coordinates, which generalise to new cities. City IDs are never used directly.
Synthetic coordinates	Coordinates are shifted from real-world values but consistent per city.	Used as relative features only; recorded distance is used for pricing.

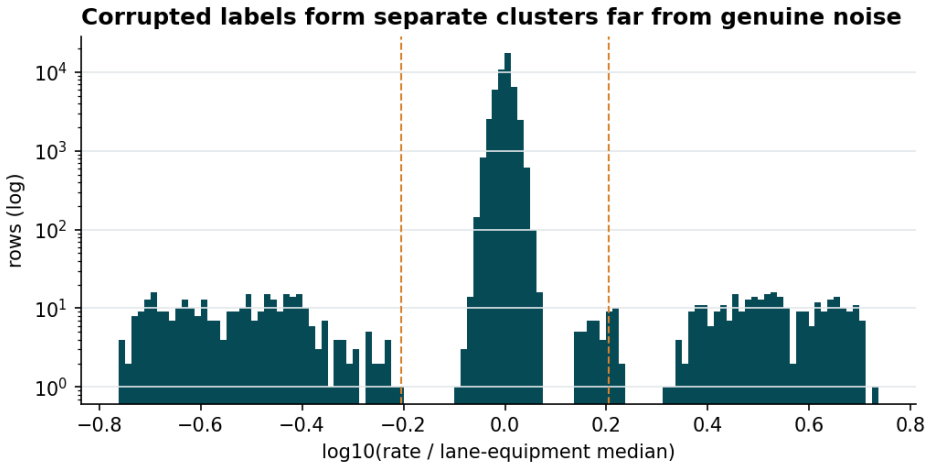


Figure 1. Ratio of each rate to its lane-equipment median (log scale). Dashed lines mark the 1.6x filter. The outer clusters are corrupted labels.

3. Key findings and model

- **Distance dominates.** log(rate) vs log(distance) correlation 0.97, with a taper: rate per mile falls on longer hauls.
- **market_index is the daily market level.** A log-linear fit gives an elasticity of about 0.15, and it has a strong weekly cycle that peaks on Thursdays.
- **Upward drift of about 0.6% a month** that market_index does not explain (Figure 2: from August to October, rate per mile rises while market_index falls).
- **Equipment and weight:** Reefer about +12% and Flatbed about +8% over Dry Van; heavier loads cost up to about 6% more.

- **Stable lane premiums of about 5%.** Lane-equipment residuals correlate 0.72 between the first and second half of the year, and 88% of validation loads run on a lane seen in training.

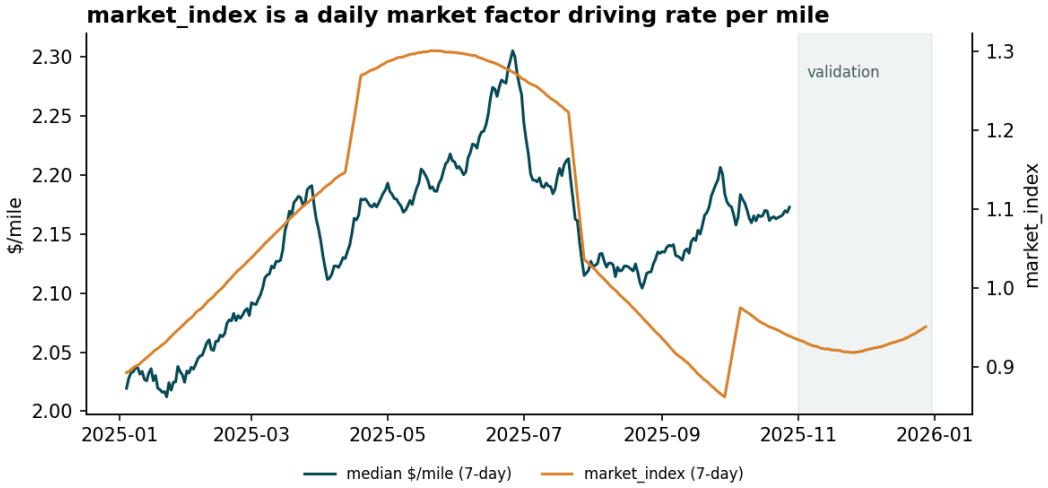


Figure 2. Median rate per mile and market_index (7-day averages). The shaded area is the prediction period.

Model. The model has three stages, all fitted on log(rate), so errors are relative and the pricing structure is multiplicative:

- **Stage 1, structural log-linear model:** log distance (and its square), equipment, log market_index, log weight, quote_signal and a **linear time trend**. This is the only stage that sees time, so the drift extrapolates into Nov-Dec. Tree models cannot extrapolate and would freeze the level at October.
- **Stage 2, lane encodings:** shrunken (k=5), out-of-fold means of the stage-1 residual by lane x equipment, lane, pickup/delivery x equipment and city, plus the support counts.
- **Stage 3, gradient boosting** (scikit-learn HistGradientBoosting) on the stage-1 residual, using distance, equipment, weight, market signals, coordinates, circuity, weekday and the lane encodings. Final prediction = exp(stage 1 + stage 3).

4. Backtest results

Model	Fold 1 Jul-Aug	Fold 2 Aug-Sep	Fold 3 Sep-Oct	Mean MAPE	Mean MAE	Mean RMSE	MAPE, all rows
Rate-per-mile baseline	3.80%	3.67%	3.68%	3.71%	$80	$111	6.04%
Plain gradient boosting	3.14%	3.55%	3.59%	3.43%	$79	$114	5.69%
Stage 1 only (log-linear + trend)	3.79%	2.80%	2.84%	3.14%	$71	$101	5.50%
Full model	3.19%	2.01%	1.73%	2.31%	$56	$86	4.66%

Per-fold MAPE and mean metrics on correctly labeled test loads. The last column includes the corrupted labels.

- Each stage adds accuracy: the lane encodings and trees cut stage-1 error by about a quarter, and the full model beats plain gradient boosting by about a third on average and by half on the most recent fold.
- Fold 3 is closest to the final setting (most training history, test right after) and reaches 1.73% MAPE. That is near the noise floor: residual noise within a lane-equipment pair is about 2.4% (std of log rate).
- Fold 1 is the weakest. With only Jan-Jun to learn from, market_index and time rise together, so the trend is overestimated (1.07% a month vs 0.53-0.66% in later folds and in the final fit). With the full Jan-Oct history, market_index rises and falls, which separates the two effects.

- On all rows, every model's RMSE is about $620 because the corrupted labels dominate it. This is the expected level for the Nov-Dec scoring if those labels contain a similar share of errors.

5. December prediction chart

The chart below was produced by the provided score.py. Lane, distance, equipment and weight are fixed, and only the date changes. Because december_chart_inputs.csv has no coordinates or market signals, the coordinates come from the city lookup, and market_index and quote_signal take each day's mean and median across the December loads in validation.csv. Predictions range from $837 to $858. That is consistent with the Sep-Oct Lexington to Fort Wayne Dry Van loads in the training data ($778-$865 for 351-371 miles). They follow the daily market level (correlation 0.94 with that day's market_index, including its Thursday peaks) plus a slight upward drift from the trend.

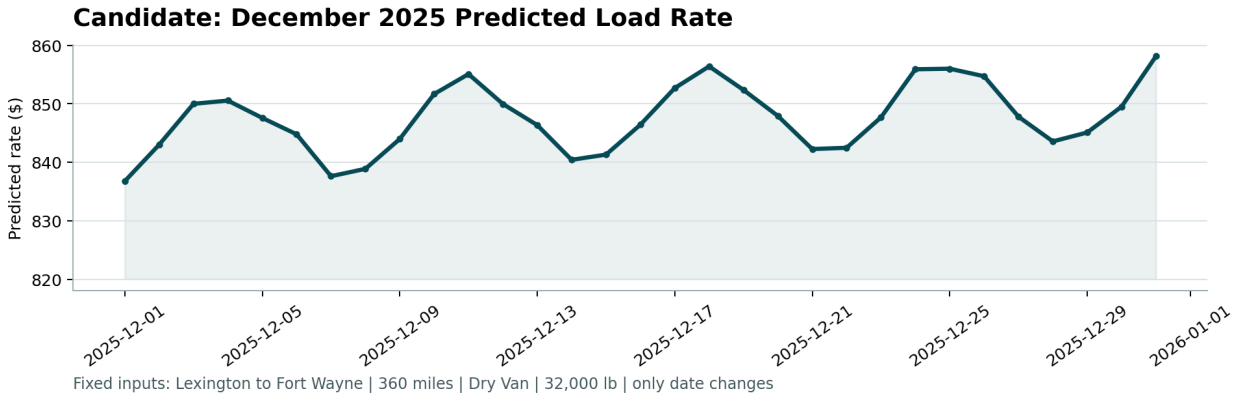


Figure 3. Fixed December prediction chart produced by score.py.

6. Limitations

- No training data covers Thanksgiving or Christmas. Holidays in the training period showed no measurable effect, so none is modeled, but late-December demand shifts could be missed.
- The linear trend is extrapolated one to two months ahead. A longer horizon would need a trend estimate that is re-fitted as new data arrives.
- Corrupted labels are excluded from training rather than corrected. If the scoring labels contain them, they set a floor on all-rows error that no model can remove.

Reproducing

```
pip install -r requirements.txt; python -m src.eda; python -m src.train; python -m src.predict; python score.py --predictions validation_predictions.csv --december-predictions data/december_chart_inputs.csv; python -m src.report
```